Saydul Akbar Murad | *Graduate Assistant*

Temporary: 313 N 37th Ave, Apt 95, MS 39401, USA.
Permanent: Dokkhin Khil Chadok, Koiar Bil, Chakaria, Cox's Bazar, Bangladesh.
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Education

2023 – Present **Ph.D. Candidate in Computer Science**
University of Southern Mississippi, Mississippi, USA.
Research Topic: *Efficient Real-Time EEG-to-Text Decoding Using LDEPTH-Based P2P Fog Computing Architecture for Scalable Brain-Computer Interfaces.* Code

2021 – 2023 **Master of Science in System Network and Security (By Research)**
Universiti Malaysia Pahang Al-Sultan Abdullah, Pahang, Malaysia
Research Topic: *A New Priority Rule Cloud Scheduling Technique That Utilizes Gaps to Increase the Efficiency of Jobs Distribution.* Code
Milestone: *Best Publication Award.*

2016 – 2020 **Bachelor of Science in Information & Communication Engineering**
Noakhali Science and Technology University, Noakhali, Bangladesh.
Project: *An Automated System to Maintain Departmental Activities Over the Cloud: An Android Application Approach.* Code

Research Interests

• EEG Signal Decoding • HPC • ML • DL • Cybersecurity • P2P Communication • Cloud Computing

Publications

*Google Scholar statistics: Total of **813** citations as of September 17, 2025, with h-index = 13, i10-index = 16*
*ResearchGate statistics: Total of **718** citations as of September 17, 2025, with h-index = 13*

Selected Journal Articles

1. **Saydul Akbar Murad**, Ashim Dahal, Nick Rahimi "*EEG-to-Text Translation: A Model for Deciphering Human Brain Activity*", arXiv, 2025
2. **Saydul Akbar Murad**, Ashim Dahal, Nick Rahimi "*Multi-Lingual Cyber Threat Detection in Tweets/X Using ML, DL, and LLM: A Comparative Analysis*", arXiv, 2025
3. Ashim Dahal, Ankit Ghimire, **Saydul Akbar Murad**, Nick Rahimi "*Redemption Score: An Evaluation Framework to Rank Image Captions While Redeeming Image Semantics and Language Pragmatics*", arXiv, 2025
4. Ashim Dahal, **Saydul Akbar Murad**, Nick Rahimi "*Heuristical Comparison of Vision Transformers Against Convolutional Neural Networks for Semantic Segmentation on Remote Sensing Imagery*", IEEE Sensors Journal, 2025
5. **Saydul Akbar Murad,** Nick Rahimi *"Unveiling Thoughts: A Review of Advancements in EEG Brain Signal Decoding into Text*", IEEE Transactions on Cognitive and Developmental Systems, 2024
6. **Saydul Akbar Murad,** Nick Rahimi *"Secure and Efficient Hierarchical P2P Fog Architecture: A Novel Approach for IoT*", IEEE Internet of Things Journal, 2024

7. **Saydul Akbar Murad,** Zafril Rizal M Azmi, Abu Jafar Md Muzahid, MD Khairul Bashar Bhuiyan, Md Saib, Nick Rahimi *"SG-PBFS: Shortest Gap-Priority Based Fair Scheduling technique for job scheduling in cloud environment*", Future Generation Computer Systems, 2024
8. **Saydul Akbar Murad,** Zafril Rizal M Azmi, Abu Jafar Md Muzahid, Md Murad Hossain Sarker, M Saef Ullah Miah, MD Khairul Bashar Bhuiyan, Nick Rahimi, Anupam Kumar Bairagi *"Priority based job scheduling technique that utilizes gaps to increase the efficiency of job distribution in cloud computing*", Sustainable Computing: Informatics and Systems, 2024
9. Nusrat Jahan Prottasha, **Saydul Akbar Murad,** Abu Jafar Md Muzahid and Md. Kowsher *"A training-based algorithm for machine learning: Impact Learning"*, Journal of Computational Science, 2023
10. Abu Jafar Md Muzahid, Syafiq Fauzi Bin Kamarulzaman, Dr Md Arafatur Rahman, **Saydul Akbar Murad** *"Multiple Vehicle Cooperation and Collision Avoidance in Automated Vehicles: Survey and an AI-Enabled Conceptual Framework"*, Scientific Reports, 2023
11. Israt Jahan, K. M. Aslam Uddin, **Saydul Akbar Murad,** M. Saef Ullah Miah, Tanvir Zaman Khan, Mehedi Masud, Sultan Aljahdali, Anupam Kumar Bairagi *"4D: a real time driver drowsiness detector using deep learning*". Electronics, 2023
12. **Saydul Akbar Murad,** Abu Jafar Md Muzahid, and Zafril Rizal Bin M Azmi *"A Review on Job Scheduling Technique in Cloud Computing and Priority Rule Based Intelligent Framework*", JKSU - Computer and Information Sciences, 2022
13. **Saydul Akbar Murad,** ABU JAFAR MD MUZAHID and Apurba Adhikary *"AI Powered Asthma Prediction Towards Treatment Formulation: An Android App Approach"*, Intelligent Automation & Soft Computing, 2022
14. Apurba Adhikary, Joy Bhuiya, **Saydul Akbar Murad,** Md. Bipul Hossain, K. M. Aslam Uddin,MD Estihad Faysal, Abidur Rahaman, Anupam Kumar Bairagi *"Performance Evaluation of Micro Lens Arrays: Improvement of Light Intensity and Efficiency of White Organic Light-Emitting Diodes"*., PLOS ONE, 2022
15. Nusrat Jahan Prottasha, Abdullah Sami, MD Kowsher, **Saydul Akbar Murad,** Anupam Kumar Bairagi, Mehedi Masud, and Mohammed Baz *"Transfer Learning for Sentiment Analysis Using BERT Base Supervised Fine Tuning"*, Sensors, 2022

Selected Conference Proceedings

1. Ashim Dahal, **Saydul Akbar Murad**, Nick Rahimi "*Embedding Shift Dissection on CLIP: Effects of Augmentations on VLM's Representation Learning*", CVPR, 2025
2. Tomas Nader, **Sadyul Akbar Murad** and Nick Rahimi, "*Human Activity Recognition using an Ensemble Learning Approach*", CATA, 2025 **(Best Paper Award)**
3. Sindhuja Penchala, **Saydul Akbar Murad**, Indranil Roy, Bidyut Gupta, Nick Rahimi, "*Unveiling text mining potential: a comparative analysis of document classification algorithms*", CATA, 2024
4. **Saydul Akbar Murad**, Nick Rahimi *"Secure and Scalable Permissioned Blockchain using LDE-P2P Networks"*, IOTSMS, 2023
5. **Saydul Akbar Murad**, Zafril Rizal M Azmi, Faria Jerin Brishti, Md Saib, Anupam Kumar Bairagi *"Priority Based Fair Scheduling: Enhancing Efficiency in Cloud Job Distribution"*, ICSECS, 2023
6. **Saydul Akbar Murad**, Nick Rahimi, Indranil Roy, Bidyut Gupta *"Design of a New Secured Hierarchical Peer-to-Peer Fog Architecture Based on Linear Diophantine Equation"*, CATA, 2023
7. Apurba Adhikary, **Saydul Akbar Murad**, Md. Shirajum Munir, and Choong Seon Hong *"Edge Assisted Crime Prediction and Evaluation Framework for Machine Learning Algorithms"*, ICOIN, 2022

8. **Saydul Akbar Murad,** Zafril Rizal Bin M Azmi, Nusrat Jahan Prottasha, and Md Kowsher *"Computer-aided system for extending the performance of diabetes analysis and prediction"*, ICSECS-ICOCSIM, 2021
9. **Saydul Akbar Murad,** Zafril Rizal M Azmi, Abu Jafar Md Muzahid, Md. Al-Imran. *"Comparative Study on Job Scheduling Using Priority Rule and Machine Learning"*, ETCCE, 2021
10. Md. Kowsher, Anik Tahabilder, **Saydul Akbar Murad** *"Impact learning: A Robust machine learning algorithm"*, ICCCM, 2021 **(Best Paper Award)**

Books and Chapters

1. **Saydul Akbar Murad**, Nick Rahimi, Indranil Roy, Bidyut Gupta "*Security challenges in the Industrial Internet of Things (IIoT)*", Quality Assessment and Security in Industrial Internet of Things, 2024
2. Nick Rahimi, **Saydul Akbar Murad**, Sarah B Lee *"Entrepreneurship Opportunities in Cybersecurity"*, Generating Entrepreneurial Ideas With AI, 2024
3. AKM Bahalul Haque, Tasfia Nausheen, Abdullah Al Mahfuj Shaan, **Saydul Akbar Murad** *"Security Attacks and Countermeasures in 5G Enabled Internet of Things"*, 5G and Beyond, 2023

Computing Skills

- **ML:** Pytorch, Transformers, Matplotlib, Tensorflow, Keras, Sklearn, NLTK, Pandas, Numpy, Scipy.
- **Simulator:** OMNeT++, CloudSim, iFogSim.
- **Programming language:** C, C++, Java, Python.
- **Database:** MySQL.
- **IDE:** Spyder, Colab, Jupyter Notebook, Code Blocks, Dev-C++, PyCharm.
- **SWE:** Git, Data Structure & Algorithm, HTML, CSS.
- Have strong knowledge on Android app development with Java.
- Have a good Knowledge on P2P communication.
- **Others:** Academic Research, Teaching, Competitive Programming.

Teaching Experience

Instructor in the Department of Computing Science at USM (Jan 2025 - Present)
✓ CSC 102: Computer Science II
✓ CSC 327: Secure Software Development

Graduate Teaching Assistant in the Department of Computing Science at USM (Jan 2023 - Dec 2023)
✓ CSC 428-591: Machine Learning in Cybersecurity
✓ Mentoring undergraduate and graduate students at the Cyber Innovation Lab (CIL).

Lecturer of Electronic & Telecom Eng. at International Islamic University Chittagong (Aug 2022 - Dec 2022)
✓ ETE–4757: Database Management Systems
✓ ETE-4758: Database Management System Sessional

Teaching Assistant in the Faculty of Computing at Universiti Malaysia Pahang (Sep 2021 - Oct 2022)
✓ BCS-3283: Mobile Application Development

Lecturer of National Model College, Noakhali, Bangladesh (Feb 2020 – Jan 2021)
✓ Data Communication
✓ Database Management Systems
✓ Programming

Academic Funding Appreciation & Awards

- **Best Paper Award**, 40th International Conference on Computers and Their Applications. 2025
- **Graduate School's Travel Award**, School of BEES & CAS, USM, USA Amount $500 2025
- **Best Proposal Project**, The University of Southern Mississippi, USA, Amount $500 2023
- **Best Researcher Award**, CAD Awards. 2023
- **Kaneka Production**, supported by Padanan Industry, Malaysia. Grant amount: RM24000. 2023
- **Best Publication Award**, University Malaysia Pahang 2023
- **Master Research Scheme (MRS)**, funded by the MHE, Malaysia. Grant amount: RM21600. 2022
- **FRGS GRANTS**, supported by the Ministry of Higher Education, Malaysia. Grant amount: RM84000. 2020
- **Best Research Paper Award**, ICCCM, ACM. 2020
- **3rd Place at NSTU Intra University Programming Contest**. 2016

Journal Editorship

- Journal of Combinatorial Mathematics and Combinatorial Computing (JCMCC)
  Publisher: Combinatorial Press
- Journal of Computer Science
  Publisher: Science Publication
- Program Committee
  34th SEDE Conference: Software & Data Engineering (SEDE)

Job Experience

Jan 2023 – Present **Graduate Research Assistant**
University of Southern Mississippi, Mississippi, USA.

May 2021 – Jan 2023 **Graduate Research Assistant**
Universiti Malaysia Pahang Al-Sultan Abdullah, Pahang, Malaysia

Aug 2022 – Dec 2022 **Lecturer**
International Islamic University Chittagong, Chittagong, Bangladesh

Feb 2020 – Jan 2021 **Lecturer**
National Model College, Noakhali, Bangladesh

Feb 2019 – Mar 2019 **Industrial Trainee**
BTCL Nandankanon, Chittagong, Bangladesh

Selected Research & Project

- **EEG to Text Decoding**
  Developed the R1 Translator model, an advanced EEG-to-text decoding framework using LSTM-based encoding and pretrained BART transformer for superior performance. Achieved SOTA results on the ZuCo dataset, surpassing existing models in key translation metrics like ROUGE, BLEU, and CER. GitHub

- **P2P Communication in Fog Computing**
  This project introduces a new integrated peer-to-peer (P2P) architecture within the fog layer. The proposed model is an advanced iteration of the existing 2-layer LDEPTH architecture. In this sophisticated framework, networks are meticulously organized in a hierarchical manner. GitHub

- **Job Scheduling Technique in Cloud Computing**
  In this project, a novel priority-based job scheduling method, designated as SG-PBFS, has been introduced and meticulously developed. Several experiments have been conducted to validate and refine this technique. GitHub

- **Department Management System Through an Android App**
  This project involved developing an Android application crafted meticulously using Java. The application is designed to serve as a comprehensive educational management tool, encompassing many functionalities. It allows for the efficient addition and management of various academic elements. GitHub

- **Impact learning Implementation**
  Impact Learning is a robust ML algorithm that is widely used to achieve more advanced results on many ML-related challenges. Impact learning is a supervised ML algorithm for resolving classification and regression problems. To install in an ML environment, just command **pip installs impact learning**

Workshops and Presentation

- **Poster Presentation**, at IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR), 2025
- **Keynote Speaker** of an online academic session on 'Publication and journal selection' of the "Workshop on Research & Publications" at UniV, Bangladesh in 2024.
- **Oral Presentation** , at The University of Southern Mississippi, USA on The Topic of Decoding Brainwaves: A Hybrid Neural Network Approach for Translating EEG Signals into Text in 2024.
- **Oral Presentation**, at 2023 10th International Conference on Internet of Things: Systems, Management and Security (IOTSMS).
- **Oral Presentation**, at 8th International Conference on Software Engineering & Computer Systems 2023.
- **Oral Presentation**, at 38th International Conference on Computers and Their Applications, USA.
- **Oral Presentation**, at National Conference for Postgraduate Research (NCON-PGR), 2022, UMP, Malaysia.
- **Oral Presentation**, at 2021 Emerging Technology in Computing, Communication, and Electronics (ETCCE), Dhaka, Bangladesh.
- **Oral Presentation**, at 7th INTERNATIONAL CONFERENCE ON SOFTWARE ENGINEERING COMPUTER SYSTEM (ICSECS – 2021), UMP, Pahang, Malaysia.

Review Research Article

■ NeurIPS ■ IEEE IoT ■ IEEE Access ■ IEEE TCDS ■ Scientific Reports ■ Computer Methods in Biomechanics and Biomedical Engineering ■ The Journal of Supercomputing ■ Innovations in Systems and Software Engineering ■ Journal of Experimental & Theoretical Artificial Intelligence ■ Congress on Smart Computing Technologies ■ Journal on Artificial Intelligence

Professional Training & Grant Writing

- NSF Grant Proposal Writing, USM (2025)
- Grant Writing Workshop, Graduate School, USM (2024)
- High Impact Publication Webinar, UMP (2022)
- Research writing skill, EMK Center (2021)
- Data Science with Python (Part 1), Maker Lab, EMK Center (2020)
- Data Science with Python (Part 2), Maker Lab, EMK Center (2020)
- Android Application Development- Java, SEIP under Basis Institute of Technology and Management (2019)